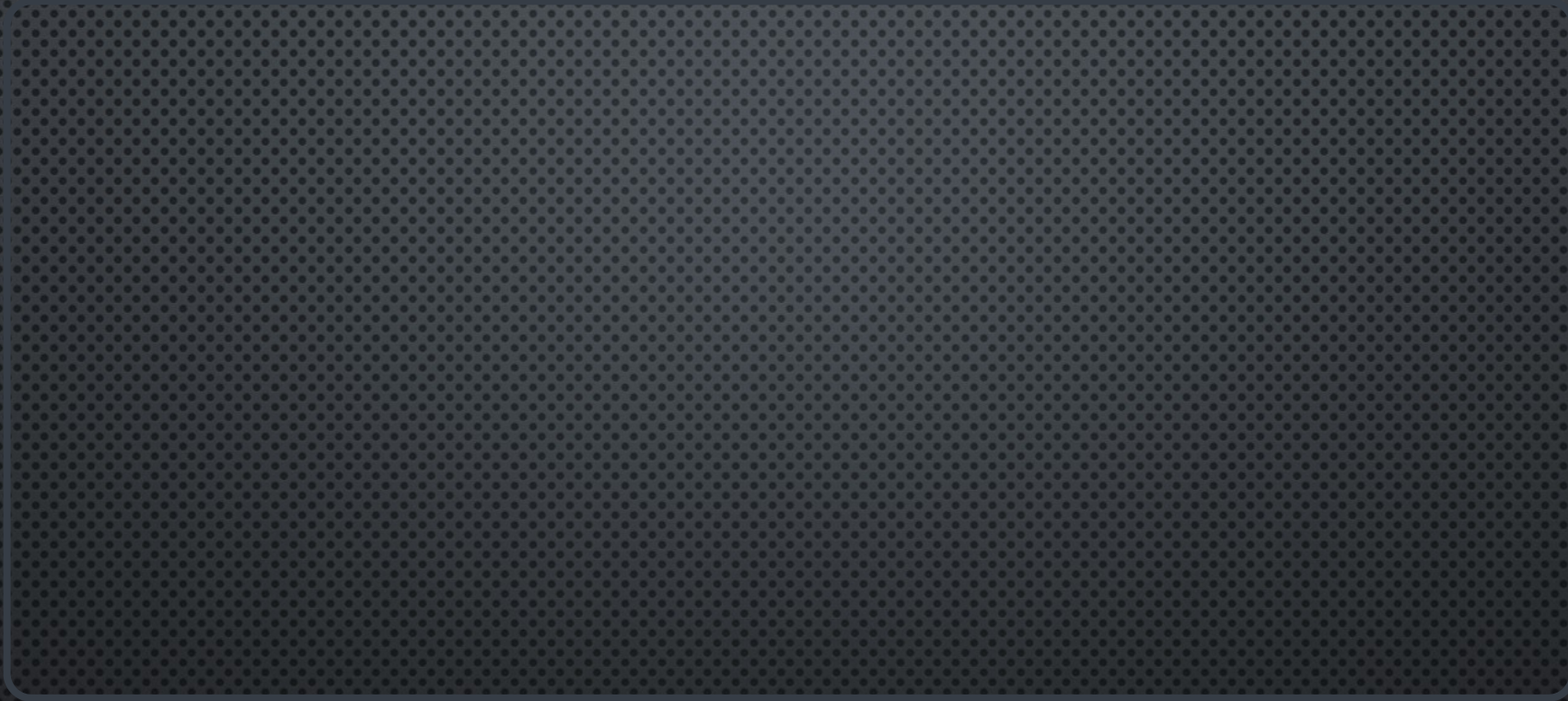




MINE SWEEPER

CHRISTIAN SALAZAR | JONATHAN CLARK | COLTON STEVENS | MUZE
MAKIESE

PROBLEM STATEMENT

*Mine fields are dangerous to clear with human hands. Even more so in forest-dense areas.

*Mine Sweeper eliminates personnel danger with a combination of autonomous and manual steering, sweeping the area and deactivating mines without the need for human hands.

*Mine Sweeper has a driver navigating through a GUI to make sure we don't miss a single mine in the process.



Problem Sketch

- Main Path
- Bonus mines

TEST FIELD SKETCH

*Test field imitates a dense forest area where a known minefield has been sectioned off.

*Mine Sweeper gets placed in a starting position and proceeds to do a sweeping motion as it traverses the field.

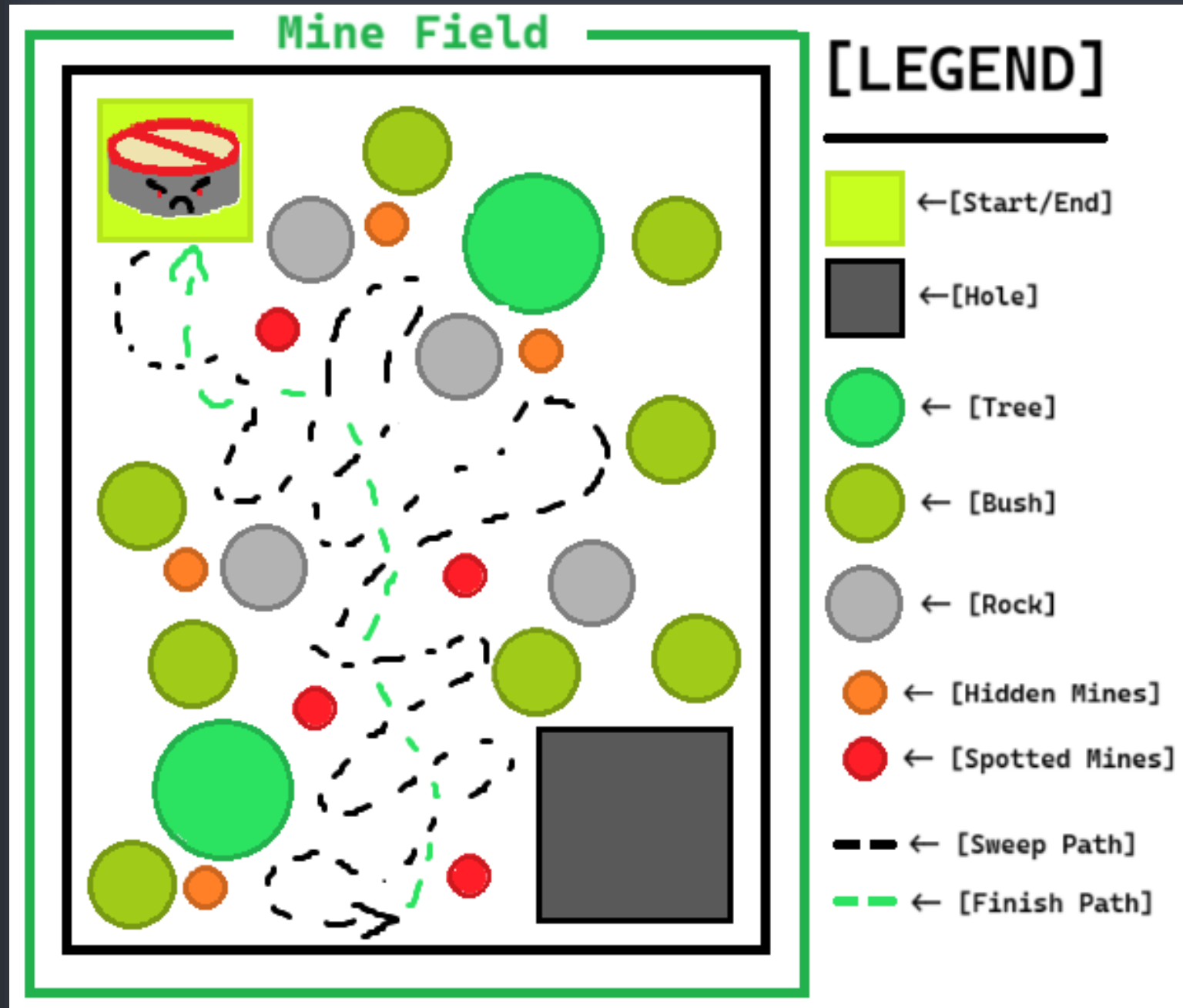
*While sweeping, Mine Sweeper must be careful not to bump into the mines or else catastrophe will happen.

*Notice there are 2 variants of mines.

"Spotted mines" indicate mines along a clear walking path.

"Hidden Mines" indicate mines outside of plain view (i.e. placed near shrubbery or in a group of rocks).

*Once Mine Sweeper has cleared the minefield, it will make it's return back to where it started to indicate a finished job and will be ready for pickup.



Technical Sketch

